Sukhrobbek Ilyosbekov

Computer Vision · Deep Learning · Explainable AI

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Research Interests

Computer Vision, Deep Learning, Medical Image Analysis, Explainable AI, Object Detection and Segmentation

Education

Northeastern University, Boston, MA *January 2024 – May 2026*

- Master of Science in Computer Science; GPA: 4.0/4.0
- Relevant Coursework: CS 7180 Advanced Perception, Machine Learning, Computer Vision

Suffolk University, Boston, MA *September 2021 – May 2023*

- Bachelor of Science in Computer Science; Cum Laude; GPA: 3.5/4.0

INTI International College Subang, Malaysia *September 2019 – July 2021*

- Bachelor of Science in Computer Science (Transfer Program)

Tashkent University of Information Technologies, Uzbekistan *September 2017 – June 2019*

- Bachelor of Science in Software Engineering

Publications

[1] S. Ilyosbekov, "MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification," *arXiv preprint*, arXiv:2512.09289, 2024. [Paper]

Awards & Honors

- **National Physics Olympiad, 1st Place,** Uzbekistan; the first student from my high school to win at the national level
- **University Coding Competition Winner,** Tashkent University of Information Technologies
- **Outstanding Project Award,** Bookshelf Scanner, Computer Vision Course, Northeastern University

Technical Skills

Deep Learning & CV: PyTorch, TensorFlow/Keras, OpenCV, YOLO, EfficientNet, ResNet, U-Net, GradCAM++, CNNs, Vision Transformers, Image Classification, Object Detection, Depth Estimation, Super-Resolution, Explainable AI

Machine Learning: Supervised/Unsupervised Learning, Transfer Learning, XGBoost, K-means Clustering, Model Evaluation, Hyperparameter Tuning, NLP

Languages: Python, C#, C++, TypeScript, JavaScript, Kotlin

Infrastructure: Docker, Kubernetes, AWS, Azure, CUDA

Frameworks & Tools: .NET, Blazor, SignalR, Prisma, NestJS, ElysiaJS, Next.js, Angular, FastAPI, RabbitMQ, PostgreSQL

Research Projects

MelanomaNet: Explainable Deep Learning for Multi-Class Skin Lesion Classification GitHub | Paper

- Developed an explainable deep learning system for multi-class skin lesion classification across all 9 ISIC 2019 diagnostic categories using EfficientNet V2 with high-resolution (384×384) input; achieved 85.61% accuracy and 0.8564 weighted F1 on 25,331 dermoscopic images.

- Implemented GradCAM++ attention visualization and novel ABCDE criterion analysis with automated feature extraction (asymmetry quantification, border irregularity, color variation via K-means, diameter calculation).
- Designed GradCAM-ABCDE alignment metrics to validate model attention correlation with clinical dermatological features, producing comprehensive interpretability reports.
- **Stack:** Python, PyTorch, EfficientNet V2, GradCAM++, OpenCV, Focal Loss, ISIC 2019 Dataset

LightDepth: Lightweight Monocular Depth Estimation GitHub

- Designed a lightweight depth estimation model using ResNet18 encoder with U-Net decoder and skip connections, achieving competitive accuracy with 42% fewer parameters (14.3M vs 24.8M) than Depth Anything V2.
- Achieved 72% faster inference while outperforming on relative error metrics (5.3% better AbsRel, 31.1% better SqRel) on NYU Depth V2.
- **Stack:** Python, PyTorch, ResNet18, U-Net Architecture, L1 Loss, NYU Depth V2 Dataset

FSRCNN: Accelerating Super-Resolution Convolutional Neural Network GitHub

- Implemented FSRCNN for single-image super-resolution (Dong et al., ECCV 2016), achieving 40× speedup over SRCNN with end-to-end learned upsampling at 2×, 3×, and 4× scales.
- Evaluated on Set5 (+1.78 dB PSNR, +0.0507 SSIM) and Set14 (+1.26 dB PSNR, +0.0477 SSIM) with ablation studies.
- **Stack:** Python, PyTorch, Mixed Precision Training (AMP), T91/Set5/Set14/DIV2K Datasets

Bookshelf Scanner: Multi-Modal Book Detection and Recognition GitHub

- Built an end-to-end pipeline combining YOLO segmentation for book instance detection with Moondream2 vision-language model for title/author extraction from bookshelf images.
- Integrated object segmentation with LLMs for multi-modal understanding and text recognition.
- **Stack:** Python, YOLO, Moondream2 LLM, Llama CPP, PyTorch, FastAPI, Angular

AoE4 Matchmaking: ML-Assisted Player Matching System GitHub

- Designed ML-assisted matchmaking combining unsupervised player clustering, XGBoost outcome prediction, and ELO rating for balanced competitive matches.
- Applied feature engineering on player statistics and ensemble methods to optimize match pairing.
- **Stack:** Python, XGBoost, K-means Clustering, Pandas, NumPy, FastAPI, Angular

Work Experience

Senior AI/ML Engineer | EmTech Care Labs | Portland, ME *Jan 2025 – Jun 2025*

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform, including a risk model that flags patients showing early signs of decline from daily activity and care-log data so caregivers can step in sooner.
- Added an NLP pipeline that reads free-text clinical notes and extracts structured details (symptoms, medications, care events), and stood up the model-serving and monitoring infrastructure under HIPAA constraints.

Senior Machine Learning Engineer | AllFactors | Santa Clara, CA *July 2022 – Dec 2023*

- Built property-valuation and market-forecasting models on real-estate data that powered the product's analytics, turning raw pricing and inventory feeds into predicted values and trend signals.
- Designed the real-time feature pipeline that fed live market data to the models and pushed updated predictions to users, then productionized the models into the customer-facing dashboards.

Software Engineer | Smart Meal Service | Russia *Sep 2020 – Oct 2021*

- Built the software for a self-service food kiosk and robotic cashier in C# / .NET, plus the customer ordering app and admin dashboard, with automated order routing that cut kitchen wait times.

Game Developer | Pentalight Technology | Malaysia *Mar 2020 – Feb 2021*

- Built the multiplayer networking for a VR smart-city simulation in Unity, keeping many users in sync in real time, plus spatial audio, hand-tracking, and a desktop spectator client.

Earlier: Freelance Developer at Freelancer.com (2019 to 2020), building Python NLP and .NET applications. Game

Developer at EC Dev Team, Uzbekistan (2016 to 2019), led RTS game development in the Clausewitz Engine.

Selected Software Projects

LogisticsX https://logisticsx.app

- Multi-tenant fleet-management platform with an AI dispatcher (Claude API, custom tool-use agent) that matches freight loads to trucks, checks federal driver hours-of-service rules, and plans routes; four web portals plus a cross-platform driver app on a .NET backend, integrated with the major freight load boards and tracking providers.

Meat.gg https://meat.gg

- Counter-Strike 2 community platform serving 30K+ users with social features and a Stripe-backed cosmetics shop, paired with a native C++ game-server plugin that bridges live game servers to the web backend for in-game moderation in real time.

OGStack https://ogstack.dev

- SaaS that auto-generates on-brand social-share preview images for any web page from a single line of markup; a multi-model AI layer (Claude, GPT, DeepSeek) reads the page and matches its branding, with AI-generated backgrounds and a global CDN (10K+ images during beta).

DepVault https://depvault.com

- Security tool that scans a project's dependencies for known vulnerabilities across eight language ecosystems and doubles as an encrypted secrets vault, with one-time sharing and a CLI for injecting tokens into CI/CD pipelines.

Med Image Scanner https://github.com/suxrobgm/med-image-scanner

- HIPAA-compliant platform that pulls X-ray, CT, and MRI scans directly from hospital imaging systems and runs PyTorch detection models on them (pneumonia on chest X-rays, intracranial hemorrhage on head CTs), with results drawn as overlays in a real radiology viewer.